

*MA.6.GR.2.1***Benchmark**

MA.6.GR.2.1 **Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.**

Benchmark Clarifications:

Clarification 1: Instruction focuses on the relationship between the area of a rectangle and the area of a right triangle.

Clarification 2: Within this benchmark, the expectation is to know from memory a formula for the area of a triangle.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.1, MA.6.NSO.2.2, MA.6.NSO.2.3
- MA.6.AR.2.3
- MA.7.GR.1.2

Terms from the K-12 Glossary

- Algorithm
- Area
- Rectangle
- Triangle

Vertical Alignment**Previous Benchmarks**

- MA.5.GR.2.1

Next Benchmarks

- MA.912.GR.4.4

Purpose and Instructional Strategies

In previous courses, students found the area of a rectangle with fractional or decimal side lengths using visual models and formulas. In grade 6 accelerated, students will use this understanding of the area of a rectangle to derive the formula of the area of a triangle (*MTR.1.1, MTR.2.1*).

Students will decompose composite figures into triangles and quadrilaterals to find area. In future courses, students will use mathematical and real-world problems involving the area of two-dimensional figures.

- Instruction includes developing the understanding that two copies of any right triangle with the same base and height will always form a rectangle. Therefore, the triangle will have an area of half of the rectangle, $A = \frac{1}{2}(\text{base} \times \text{height})$. This understanding can develop from seeing how a triangle is constructed when cutting a rectangular piece of paper diagonally in half.
- Students should be flexible in their understanding of formulas to be able to use the equivalency of $\frac{1}{2}bh$ and $\frac{bh}{2}$.
- Formulas can be a tool or strategy for geometric reasoning. Students require a solid understanding of two area concepts: (1) the area of a rectangle is *length* $\times$ *width* or *base* $\times$ *height*, and (2) figures of the same size and shape (congruent) have the same area.
- Instruction includes representing measurements for the area as square units, units squared or units².

- Students should understand that any side of the triangle can be a base; however, the height can only be represented as a line segment drawn from a vertex perpendicular to the base. The terms height and altitude can be used interchangeably. Students should see the right-angle symbol, $\perp$ to indicate perpendicularity.
- Problem types include having students measure lengths using a ruler to determine the area.

Common Misconceptions or Errors

- Students may forget that multiplying by $\frac{1}{2}$ and dividing by 2 are the same operation.
- Students may neglect to apply the $\frac{1}{2}$ when finding the area of a triangle.
- Students may incorrectly identify a side measurement as the height of a triangle.
- Students may incorrectly measure with a ruler.

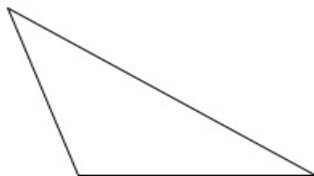
Strategies to Support Tiered Instruction

- Teacher models several problems solving them both ways (using a rectangle and using a formula). Then have the students solve them step by step with guidance. This will provide students with the opportunity to see that the two operations are identical.
- Teacher reinforces that a right triangle is half of a rectangle cut diagonally. Therefore, we must cut the area in half.
- Teacher models with geometric software so students can see that a right triangle is half of a rectangle, which is why we multiply by $\frac{1}{2}$.
- Teacher models the use of manipulatives that students can measure to better understand there is a difference between a side length and the height in non-right triangles.

Instructional Tasks

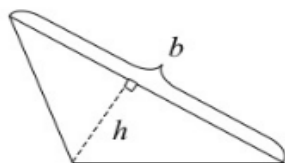
Instructional Task 1 (MTR.2.1, MTR.4.1)

Mrs. Lito asked her students to label a base b and its corresponding height h in the triangle shown.

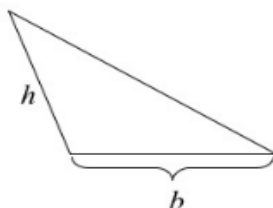


Three students drew the figures below.

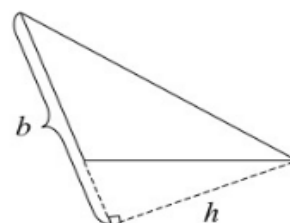
A. Raul



B. Mark



C. Kiki

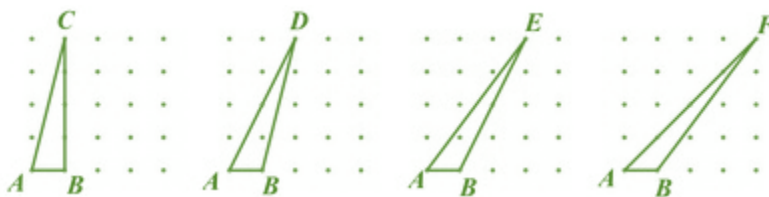


Part A. Which students, if any, have correctly identified a base and its corresponding height? Which ones have not? Explain what is incorrect.

Part B. There are three possible base-height pairs for this triangle. Sketch all three.

Instructional Task 2 (MTR.5.1)

Look at the triangles below.

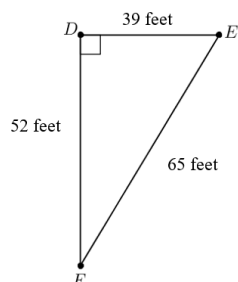


Determine and explain:

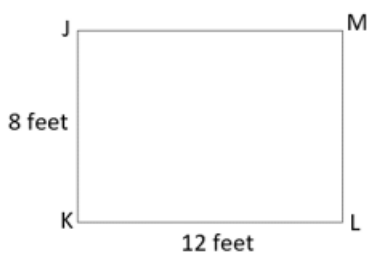
- Which triangle has the greatest area?
- Which triangle has the least area?
- Do any of the triangles have the same area?
- Are some areas impossible to compare?

Instructional Items*Instructional Item 1*

Find the area of $\triangle DEF$.

*Instructional Item 2*

Using the rectangle below, if a line was drawn from vertex J to vertex L , what is the area of $\triangle JKL$?



**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*